We create chemistry

# Safety data sheet



BASF Safety data sheet according to Regulation UK SI 2019/758 and UK SI 2020/1577 as amended from time to time.
Date / Revised: 11.06.2025 Version: 13.0
Date / Previous version: 16.10.2024 Previous version: 12.0
Product: **Potassium Metabisulfite food grade (E224)**
(ID no. 30042359/SDS_GEN_GB/EN)
Date of print 30.06.2025

## SECTION 1: Identification of the substance/mixture and of the company/undertaking

### 1.1. Product identifier

# Potassium Metabisulfite food grade (E224)

Chemical name: Dipotassium disulphite; potassium metabisulfite
CAS Number: 16731-55-8

REACH registration number: 01-2119537422-45-0001

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: food additive(s)
Recommended use: inorganic reducing agents, initial product for chemical syntheses, process chemical

For the detailed identified uses of the product see appendix of the safety data sheet.

### 1.3. Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY

Contact address:
BASF plc
4th and 5th Floors, 2 Stockport Exchange
Railway Road, Stockport, SK1 3GG
UNITED KINGDOM

Telephone: +44 161 475 3000
E-mail address: product-safety-uk-and-ireland@basf.com

### 1.4. Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

# SECTION 2: Hazards Identification

## 2.1. Classification of the substance or mixture

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

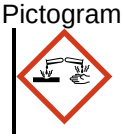
| | |
|---|---|
| Skin Irrit. 2 | H315 Causes skin irritation. |
| Eye Dam. 1 | H318 Causes serious eye damage. |

For the classifications not written out in full in this section the full text can be found in section 16.

## 2.2. Label elements

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Pictogram:



Signal Word:
Danger

Hazard Statement:

| | |
|---|---|
| H318 | Causes serious eye damage. |
| H315 | Causes skin irritation. |

Precautionary Statements (Prevention):

| | |
|---|---|
| P280 | Wear protective gloves and eye protection or face protection. |
| P264 | Wash contaminated body parts thoroughly after handling. |

Precautionary Statements (Response):

| | |
|---|---|
| P305 + P351 + P338 | IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. |
| P310 | Immediately call a POISON CENTER or physician. |
| P302 + P352 | IF ON SKIN: Wash with plenty of soap and water. |
| P362 + P364 | Take off contaminated clothing and wash it before reuse. |

Labeling of special preparations (GHS):
EUH031: Contact with acids liberates toxic gas.

Hazard determining component(s) for labelling: Dipotassium disulphite

## 2.3. Other hazards

According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

No specific dangers known, if the regulations/notes for storage and handling are considered.

The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

## SECTION 3: Composition/Information on Ingredients

### 3.1. Substances

Chemical nature

Dipotassium disulphite
CAS Number: 16731-55-8
EC-Number: 240-795-3

$K_2S_2O_5$

Hazardous ingredients (GHS)

Dipotassium disulphite

| | |
|---|---|
| Content (W/W): >= 75 % - <= 100 % | Skin Irrit. 2<br>Eye Dam. 1 |
| CAS Number: 16731-55-8<br>EC-Number: 240-795-3 | H318, H315 |

For the classifications not written out in full in this section, including the hazard classes and the hazard statements, the full text is listed in section 16.

### 3.2. Mixtures

Not applicable

## SECTION 4: First-Aid Measures

### 4.1. Description of first aid measures

Remove contaminated clothing.

If inhaled:
If difficulties occur after dust has been inhaled, remove to fresh air and seek medical attention. After inhalation of decomposition products: Immediately administer a corticosteroid from a controlled/metered dose inhaler. Seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

### 4.2. Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., Many individuals are sensitive to sulphite additives and may experience a range of symptoms, including dermatitis, urticaria, angio-oedema, abdominal pain, diarrhoea, bronchoconstriction and anaphylaxis.

Hazards: Risk of sulfur dioxide formation by reaction with gastric acid after swallowing.

### 4.3. Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

## SECTION 5: Fire-Fighting Measures

### 5.1. Extinguishing media

Suitable extinguishing media:
water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:
water jet

Additional information:
Product will not burn.

Use extinguishing measures to suit surroundings.

### 5.2. Special hazards arising from the substance or mixture

Endangering substances: sulphur dioxide
Advice: The substances/groups of substances mentioned can be released if the product is involved in a fire.

### 5.3. Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered. Contaminated extinguishing water must be disposed of in accordance with official regulations. In case of fire and/or explosion do not breathe fumes.

## SECTION 6: Accidental Release Measures

### 6.1. Personal precautions, protective equipment and emergency procedures

Avoid contact with the skin, eyes and clothing. Use personal protective clothing. Ensure adequate ventilation. Avoid dust formation.

### 6.2. Environmental precautions

Do not discharge into drains/surface waters/groundwater. Do not discharge into the subsoil/soil. Retain and dispose of contaminated wash water.

### 6.3. Methods and material for containment and cleaning up

Sweep/shovel up. Dispose of absorbed material in accordance with regulations.

### 6.4. Reference to other sections

Information regarding exposure controls/personal protection and disposal considerations can be found in section 8 and 13.

## SECTION 7: Handling and Storage

### 7.1. Precautions for safe handling

Use only in well-ventilated areas. Avoid dust formation. Avoid contact with skin and eyes.

Protection against fire and explosion:
The substance/product is non-combustible. No special precautions necessary.

### 7.2. Conditions for safe storage, including any incompatibilities

Segregate from acids and acid forming substances. Segregate from oxidants.

Suitable materials for containers: Stainless steel 1.4571, Stainless steel 1.4541, Low density polyethylene (LDPE), High density polyethylene (HDPE), Carbon steel (Iron), rubberized, Polyester resin, glass reinforced (Palatal A410)
Further information on storage conditions: Keep away from heat. Keep container tightly closed in a cool, well-ventilated place. Keep container dry. The product consumes oxygen. Danger of lack of oxygen in containers and tanks.

### 7.3. Specific end use(s)

See exposure scenario(s) in the attachment to this safety data sheet.

## SECTION 8: Exposure Controls/Personal Protection

### 8.1. Control parameters

Components with occupational exposure limits

The nuisance dust limit value is to be kept.

The substance mentioned develops if the regulation/notes for storage and handling are not observed.

7446-09-5: sulphur dioxide
TWA value 1.3 mg/m3 ; 0.5 ppm (WEL/EH 40 (UK))
STEL value 2.7 mg/m3 ; 1 ppm (WEL/EH 40 (UK))
STEL value 2.7 mg/m3 ; 1 ppm (WEL/EH 40 (UK))
Ceiling limit value/factor: 15 min
TWA value 1.3 mg/m3 ; 0.5 ppm (EU SCOEL)
Ceiling limit value/factor: 8HR
STEL value 2.7 mg/m3 ; 1.0 ppm (EU SCOEL)
Ceiling limit value/factor: 15 min

7681-57-4: sodium metabisulphite
TWA value 5 mg/m3 (WEL/EH 40 (UK))

PNEC
freshwater:
No hazard identified.

marine water:
No hazard identified.

sediment (freshwater):
No hazard identified.

sediment (marine water):
No hazard identified.

STP:
No hazard identified.

air:
No hazard identified.

soil:
No hazard identified.

DNEL
worker:
Long-term exposure- systemic effects, Inhalation: 263 mg/m3

consumer:
Long-term exposure- systemic effects, Inhalation: 78 mg/m3

consumer:
Long-term exposure- systemic effects, oral: 10 mg/kg

worker:
Inhalation
The nuisance dust limit (inhalativ fraction) was used as basis for the DNEL.

### 8.2. Exposure controls

Personal protective equipment

Respiratory protection:
Breathing protection if dusts are formed. Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1or FFP1) Breathing protection if gases/vapours are formed. Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Chemical resistant protective gloves (EN ISO 374-1)
Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):
e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other
Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing. Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:
Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
Handle in accordance with good industrial hygiene and safety practice. Do not inhale vapours or dust. Hands and/or face should be washed before breaks and at the end of the shift.

## SECTION 9: Physical and Chemical Properties

### 9.1. Information on basic physical and chemical properties

| | |
|---|---|
| Form: | powder |
| Colour: | white |
| Odour: | faint odour, of sulfur dioxide |
| Odour threshold: | Not determined due to potential health hazard by inhalation. |
| pH value: | 3.8 - 4.6<br>(water, 5 %(m), 20 °C)<br>No data available. |

| | | |
|---|---|---|
| decomposition point: | approx. 150 °C<br>Literature data. | |
| Melting point: | The substance / product decomposes therefore not determined. | |
| Boiling point: | (1,013 hPa)<br>not applicable | |
| Flash point: | not applicable, the product is a solid | |
| Evaporation rate: | The product is a non-volatile solid. | |
| Flammability: | not flammable | (other) |
| Lower explosion limit: | For solids not relevant for classification and labelling. | |
| Upper explosion limit: | For solids not relevant for classification and labelling. | |
| Vapour pressure: | not applicable | |
| Density: | 2.3 g/cm3<br>(20 °C)<br>Literature data. | |
| Relative density: | 2.3<br>(20 °C) | |
| Solubility in water: | Literature data.<br>495 g/l<br>(25 °C) | |
| Partitioning coefficient n-octanol/water (log Kow): | not applicable | |
| Self ignition: | not self-igniting | Test type: Self-ignition at high temperatures. |
| | not self-igniting | Test type: Spontaneous self-ignition at room-temperature. |
| Thermal decomposition: | > 150 °C<br>To avoid thermal decomposition, do not overheat. | |
| Viscosity, dynamic: | not applicable | |
| Viscosity, kinematic: | not applicable, the product is a solid | |
| Explosion hazard: | Based on the chemical structure there is no indication of explosive properties. | |
| Fire promoting properties: | Based on its structural properties the product is not classified as oxidizing. | |

### 9.2. Other information

| | | |
|---|---|---|
| Burning rate: | Study scientifically not justified. | |
| Self heating ability: | It is not a substance capable of spontaneous heating. | |
| Bulk density: | 1,100 - 1,300 kg/m3 | |
| pKA: | not applicable | |
| Adsorption/water - soil: | Study scientifically not justified. | |
| Surface tension: | Based on chemical structure, surface activity is not to be expected. | (other) |
| Grain size distribution | 65 µm | (D10, ISO 13320-1;; particle size by laser diffraction) |
| | 228 µm | (D50, ISO 13320-1;; particle size by laser diffraction) |
| | 523 µm | (D90, ISO 13320-1;; particle size by laser diffraction) |
| | fine particles | |

## SECTION 10: Stability and Reactivity

### 10.1. Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

### 10.2. Chemical stability
The product is stable if stored and handled as prescribed/indicated.

### 10.3. Possibility of hazardous reactions
Reacts with nitrites. Reacts with nitrates. Reacts with oxidizing agents. Generation of sulphur dioxide upon exposure to acids. (or conditions.) The product consumes oxygen.

### 10.4. Conditions to avoid
Avoid humidity. avoid atmospheric oxygen

### 10.5. Incompatible materials

Substances to avoid:
nitrites, nitrates, oxidizing agents, acids

### 10.6. Hazardous decomposition products

Hazardous decomposition products:
sulphur dioxide

# SECTION 11: Toxicological Information

## 11.1. Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of low toxicity after single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Experimental/calculated data:
LD50 rat (oral): approx. 2,300 mg/kg (similar to OECD guideline 401)

LC50 rat (by inhalation): > 5.5 mg/l 4 h (OECD Guideline 403)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. Tested as dust aerosol.

LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Irritation

Assessment of irritating effects:
Causes serious eye damage. Causes skin irritation.

Experimental/calculated data:
Skin corrosion/irritation
rabbit: non-irritant (BASF-Test)

Skin corrosion/irritation
In vitro assay: Irritant. (OECD Guideline 439)

Skin corrosion/irritation
In vitro assay: Non corrosive. (OECD Guideline 435)

Serious eye damage/irritation
rabbit: irreversible damage (OECD Guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Skin sensitizing effects were not observed in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. A sensitizing effect on particularly sensitive individuals cannot be excluded.

Experimental/calculated data:
Mouse Local Lymph Node Assay (LLNA) mouse: Non-sensitizing. (OECD Guideline 429)
The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Germ cell mutagenicity

Assessment of mutagenicity:
No mutagenic effect was found in various tests with bacteria and mammalian cell culture. The substance was not mutagenic in a test with mammals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Carcinogenicity

Assessment of carcinogenicity:
In long-term animal studies in which the substance was given in the drinking water in high doses, a carcinogenic effect was not observed.

Reproductive toxicity

Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Developmental toxicity

Assessment of teratogenicity:
No indications of a developmental toxic / teratogenic effect were seen in animal studies.

Experiences in humans

Experimental/calculated data:

With sensitive persons it can lead to an over sensitive reaction.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
No substance-specific organtoxicity was observed after repeated administration to animals. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration hazard

not applicable

## SECTION 12: Ecological Information

### 12.1. Toxicity

Assessment of aquatic toxicity:
There is a high probability that the product is not acutely harmful to aquatic organisms.

Toxicity to fish:
LC50 (96 h) 316 mg/l, Leuciscus idus (DIN 38412 Part 15, static)
Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic invertebrates:
EC50 (48 h) 89 mg/l, Daphnia magna (Directive 79/831/EEC, static)
Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

EC50 (48 h) 230 mg/l, Daphnia magna (other, static)
The details of the toxic effect relate to the nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic plants:
EC50 (72 h) 43.8 mg/l (growth rate), Scenedesmus subspicatus (Algal growth inhibition test, static)
Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

EC50 (72 h) > 100 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201, static)
Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Microorganisms/Effect on activated sludge:
No observed effect concentration (180 min) >= 1,000 mg/l, (OECD Guideline 209, aquatic)
Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Chronic toxicity to fish:
No observed effect concentration (34 d) >= 316 mg/l, Brachydanio rerio (OECD Guideline 210, Flow through.)
The details of the toxic effect relate to the nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d) > 10 mg/l, Daphnia magna (OECD Guideline 211, semistatic)
Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Assessment of terrestrial toxicity:
No data available.
Study does not need to be conducted.

## 12.2. Persistence and degradability

Assessment biodegradation and elimination (H2O):
Inorganic product which cannot be eliminated from water by biological purification processes.
Study scientifically not justified.

Elimination information:
Study scientifically not justified.

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.
Study scientifically not justified.
Information on Stability in Water (Hydrolysis):
Study scientifically not justified.

## 12.3. Bioaccumulative potential

Assessment bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

## 12.4. Mobility in soil

Assessment transport between environmental compartments:
Volatility: The substance will not evaporate into the atmosphere from the water surface. Study scientifically not justified.
Adsorption in soil: Adsorption to solid soil phase is not expected. Study scientifically not justified.

## 12.5. Results of PBT and vPvB assessment

According to Annex XIII of Regulation (EC) No.1907/2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH): The product does not fulfill the criteria for PBT (Persistent/bioaccumulative/toxic) and vPvB (very persistent/very bioaccumulative). Self classification

## 12.6. Other adverse effects

The substance is not listed in Regulation (EC) 1005/2009 on substances that deplete the ozone layer.

## 12.7. Additional information

Sum parameter

Chemical oxygen demand (COD): (calculated) approx. 140 mg/g

Other ecotoxicological advice:
Higher concentrations of the substance may cause a strong chemical oxygen consumption in biological sewage-treatment plants and/or waterways. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

## SECTION 13: Disposal Considerations

### 13.1. Waste treatment methods

The UK Environmental Protection (Duty of Care) Regulations (EP) and amendments should be noted (United Kingdom).
This product and any uncleaned containers must be disposed of as hazardous waste in accordance with the 2005 Hazardous Waste Regulations and amendments (United Kingdom)

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

## SECTION 14: Transport Information

**Land transport**

ADR

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

RID

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Inland waterway transport**

ADN

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user: | None known |

Transport in inland waterway vessel
Not evaluated

**Sea transport**

IMDG

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

**Air transport**

IATA/ICAO

| | |
|---|---|
| | Not classified as a dangerous good under transport regulations |
| UN number or ID number: | Not applicable |
| UN proper shipping name: | Not applicable |
| Transport hazard class(es): | Not applicable |
| Packing group: | Not applicable |
| Environmental hazards: | Not applicable |
| Special precautions for user | None known |

### 14.1. UN number or ID number
See corresponding entries for “UN number or ID number” for the respective regulations in the tables above.

### 14.2. UN proper shipping name
See corresponding entries for “UN proper shipping name” for the respective regulations in the tables above.

### 14.3. Transport hazard class(es)

See corresponding entries for “Transport hazard class(es)” for the respective regulations in the tables above.

### 14.4. Packing group

See corresponding entries for “Packing group” for the respective regulations in the tables above.

### 14.5. Environmental hazards

See corresponding entries for “Environmental hazards” for the respective regulations in the tables above.

### 14.6. Special precautions for user

See corresponding entries for “Special precautions for user” for the respective regulations in the tables above.

### 14.7. Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

## SECTION 15: Regulatory Information

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

The data should be considered when making any assessment under the Control of Substances Hazardous to Health Regulations (COSHH), and related guidance, for example, 'COSHH Essentials' (United Kingdom).

### 15.2. Chemical Safety Assessment

Chemical Safety Assessment performed

## SECTION 16: Other Information

Assessment of the hazard classes according to UN GHS criteria (most recent version)

Eye Dam. 1
Acute Tox. 5 (oral)
Aquatic Acute 3
Skin Irrit. 2

Any other intended applications should be discussed with the manufacturer.

Full text of the classifications, including the hazard classes and the hazard statements, if mentioned in section 2 or 3:

| | |
|---|---|
| Skin Irrit. | Skin irritation |
| Eye Dam. | Serious eye damage |
| H318 | Causes serious eye damage. |
| H315 | Causes skin irritation. |

Abbreviations
ADR = The European Agreement concerning the International Carriage of Dangerous Goods by Road. ADN = The European Agreement concerning the International Carriage of Dangerous Goods by Inland waterways. ATE = Acute Toxicity Estimates. CAO = Cargo Aircraft Only. CAS = Chemical Abstract Service. CLP = Classification, Labelling and Packaging of substances and mixtures. DIN = German national organization for standardization. DNEL = Derived No Effect Level. EC50 = Effective concentration median for 50% of the population. EC = European Community. EN = European Standards. IARC = International Agency for Research on Cancer. IATA = International Air Transport Association. IBC-Code = Intermediate Bulk Container code. IMDG = International Maritime Dangerous Goods Code. ISO = International Organization for Standardization. STEL = Short-Term Exposure Limit. LC50 = Lethal concentration median for 50% of the population. LD50 = Lethal dose median for 50% of the population. TLV = Threshold Limit Value. MARPOL = The International Convention for the Prevention of Pollution from Ships. NEN = Dutch Norm. NOEC = No Observed Effect Concentration. OEL = Occupational Exposure Limit. OECD = Organization for Economic Cooperation and Development. PBT = Persistent, Bioaccumulative and Toxic. PNEC = Predicted No Effect Level. PPM = Parts per million. RID = The European Agreement concerning the International Carriage of Dangerous Goods by Rail. TWA = Time Weight Average. UN-number = UN number at transport. vPvB = very Persistent and very Bioaccumulative.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

## Annex: Exposure Scenarios

**Index**

**1.** Formulation of fertilizer products, Agriculture, forestry, fishery
F; ERC3; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28; PC12

**2.** Formulation, Use in/as Photochemicals
F; ERC2, ERC3; PROC3, PROC4, PROC5, PROC8b, PROC9, PROC15, PROC26, PROC28

**3.** Formulation, Use in food products
F; ERC2, ERC3; PROC3, PROC4, PROC5, PROC8b, PROC9, PROC15, PROC26, PROC28

**4.** Formulation & (re)packing of substances and mixtures
F; ERC3

**5.** Use as an intermediate, (use in industrial settings)
IS; SU8, SU9; ERC6a; PROC3, PROC1, PROC15, PROC26, PROC28

**6.** Use in water treatment agents, Use in mining chemicals, Marine and Offshore, Use in Surface treatment products, Use in Metal surface treatment, (use in industrial settings)
IS; SU2a, SU2b, SU23; ERC6b; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28

**7.** Use in food products, (use in industrial settings)
IS; SU4; ERC6b; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28

**8.** Use in/as Photochemicals, (use in industrial settings)
IS; SU6b, SU7; ERC6b; PROC3, PROC4, PROC5, PROC7, PROC8b, PROC9, PROC10, PROC13, PROC15, PROC26, PROC28

**9.** Use in/as Fertilizers, Agriculture, forestry, fishery, (use in professional settings)
PW; SU1; ERC8e; PROC3, PROC4, PROC5, PROC15, PROC19, PROC21, PROC26; PC12

**10.** Use in water treatment agents, Use in mining chemicals, Marine and Offshore, Use in Surface treatment products, Use in Metal surface treatment, (use in professional settings)
PW; SU2a, SU2b, SU23; ERC8b; PROC3, PROC4, PROC5, PROC15, PROC19, PROC26

**11.** Use in food products, (use in professional settings)
PW; SU4; ERC8b; PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC10, PROC11, PROC13, PROC19, PROC26

**12.** Use in/as Photochemicals, (use in professional settings)
PW; SU6b, SU7; ERC8e; PROC3, PROC4, PROC5, PROC8b, PROC8a, PROC9, PROC10, PROC11, PROC13, PROC15, PROC19, PROC26

**13.** Use in/as Fertilizers, (liquid products), (consumer use)
C; ERC8e; PC12

**14.** Use in/as Photochemicals, (consumer use)
C; ERC8e; PC30

**15.**Formulation & (re)packing of substances and mixtures
IS; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28

**********

## 1. Short title of exposure scenario

Formulation of fertilizer products, Agriculture, forestry, fishery
F; ERC3; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28; PC12

## Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC3: Formulation into solid matrix<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to | |

| | |
|---|---|
| prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. | |

| | |
|---|---|
| Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |

| | |
|---|---|
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through | |

| | |
|---|---|
| time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular | |

| | |
|---|---|
| inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |

| | |
|---|---|
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with | |

| | |
|---|---|
| sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source | |

| | |
|---|---|
| and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid | |

| | |
|---|---|
| splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 2. Short title of exposure scenario

Formulation, Use in/as Photochemicals
F; ERC2, ERC3; PROC3, PROC4, PROC5, PROC8b, PROC9, PROC15, PROC26, PROC28

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC2: Formulation into mixture<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC3: Formulation into solid matrix<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular | |

| | |
|---|---|
| inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |

| | |
|---|---|
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with | |

| | |
|---|---|
| sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to | |

| | |
|---|---|
| points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| | Additional PROC(s) covered: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs | |

| | |
|---|---|
| followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| |
|---|
| Local exhaust ventilation and / or general ventilation are / is advisable. |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. | |

| | |
|---|---|
| Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off | |

| | |
|---|---|
| any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the | |

| | |
|---|---|
| source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear | |

| | |
|---|---|
| suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through | |

| | |
|---|---|
| time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 3. Short title of exposure scenario

Formulation, Use in food products
F; ERC2, ERC3; PROC3, PROC4, PROC5, PROC8b, PROC9, PROC15, PROC26, PROC28

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC2: Formulation into mixture<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC3: Formulation into solid matrix<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. | |

| | |
|---|---|
| Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to | |

| | |
|---|---|
| prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the | |

| | |
|---|---|
| source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear | |

| | |
|---|---|
| suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| | Additional PROC(s) covered: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to | |

| | |
|---|---|
| points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up | |

| | |
|---|---|
| contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature |

| | |
|---|---|
| | Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure | |

| | |
|---|---|
| to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general | |

| | |
|---|---|
| ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place | |

| | |
|---|---|
| are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |

| |
|---|
| Local exhaust ventilation and / or general ventilation are / is advisable. |
| Local exhaust ventilation and / or general ventilation are / is advisable. |

**********

## 4. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures
F; ERC3

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC3: Formulation into solid matrix<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

**********

## 5. Short title of exposure scenario

Use as an intermediate, (use in industrial settings)
IS; SU8, SU9; ERC6a; PROC3, PROC1, PROC15, PROC26, PROC28

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC6a: Use of intermediate<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| | Additional PROC(s) covered: Chemical production or refinery in closed process without likelihood of exposure or processes with equivalent containment conditions. |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |

| ***Risk Management Measures*** | |
|---|---|
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual | |

| | |
|---|---|
| phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and | |

| | |
|---|---|
| work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. | |

| | |
|---|---|
| Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |

| | |
|---|---|
| | |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically | |

| | |
|---|---|
| resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 6. Short title of exposure scenario

Use in water treatment agents, Use in mining chemicals, Marine and Offshore, Use in Surface treatment products, Use in Metal surface treatment, (use in industrial settings)
IS; SU2a, SU2b, SU23; ERC6b; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC6b: Use of reactive processing aid at industrial site (no inclusion into or onto article)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in | |

| | |
|---|---|
| place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |

| ***Additional good practice advice*** |
|---|
| Local exhaust ventilation and / or general ventilation are / is advisable. |
| Local exhaust ventilation and / or general ventilation are / is advisable. |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |

| | |
|---|---|
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide | |

| | |
|---|---|
| specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. | |

| | |
|---|---|
| Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |

| | |
|---|---|
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration | |

| | |
|---|---|
| of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are | |

| | |
|---|---|
| implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 7. Short title of exposure scenario

Use in food products, (use in industrial settings)

IS; SU4; ERC6b; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC6b: Use of reactive processing aid at industrial site (no inclusion into or onto article)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance | |

| | |
|---|---|
| within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee | |

| | |
|---|---|
| training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination | |

| | |
|---|---|
| immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure | |

| | |
|---|---|
| containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with | |

| | |
|---|---|
| contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |

| | |
|---|---|
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 8. Short title of exposure scenario

Use in/as Photochemicals, (use in industrial settings)
IS; SU6b, SU7; ERC6b; PROC3, PROC4, PROC5, PROC7, PROC8b, PROC9, PROC10, PROC13, PROC15, PROC26, PROC28

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC6b: Use of reactive processing aid at industrial site (no inclusion into or onto article)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable | |

| | |
|---|---|
| chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure | |

| | |
|---|---|
| containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid | |

| | |
|---|---|
| splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| |
|---|
| **Contributing exposure scenario** |

| | |
|---|---|
| **Use descriptors covered** | PROC7: Industrial spraying<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear | |

| | |
|---|---|
| suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: industrial |
| | Additional PROC(s) covered: Transfer of substance or preparation into small containers (dedicated filling line, including weighing). |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to | |

| | |
|---|---|
| prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the | |

| | |
|---|---|
| source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear | |

| | |
|---|---|
| suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid | |

| | |
|---|---|
| inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of | |

| | |
|---|---|
| equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance | 0.0001 Pa |

| | |
|---|---|
| during use | |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |

| | |
|---|---|
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to | |

| | |
|---|---|
| points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 9. Short title of exposure scenario

Use in/as Fertilizers, Agriculture, forestry, fishery, (use in professional settings)
PW; SU1; ERC8e; PROC3, PROC4, PROC5, PROC15, PROC19, PROC21, PROC26; PC12

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8e: Widespread use of reactive processing aid (no inclusion into or onto article, outdoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional |

| | |
|---|---|
| | controlled exposure or processes with equivalent containment condition<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure | |

| | |
|---|---|
| containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in | |

| | |
|---|---|
| place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |

| ***Additional good practice advice*** |
|---|
| Local exhaust ventilation and / or general ventilation are / is advisable. |
| Local exhaust ventilation and / or general ventilation are / is advisable. |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |

| | |
|---|---|
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC19: Manual activities involving hand contact<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to | |

| | |
|---|---|
| prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC21: Low energy manipulation and handling of substances bound in/on materials or articles<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | Solid, low dustiness |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. | |

| | |
|---|---|
| Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |

| | |
|---|---|
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration | |

| | |
|---|---|
| of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 10. Short title of exposure scenario

Use in water treatment agents, Use in mining chemicals, Marine and Offshore, Use in Surface treatment products, Use in Metal surface treatment, (use in professional settings)
PW; SU2a, SU2b, SU23; ERC8b; PROC3, PROC4, PROC5, PROC15, PROC19, PROC26

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8b: Widespread use of reactive processing aid (no inclusion into or onto article, indoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition |

| | |
|---|---|
| | Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure | |

| | |
|---|---|
| to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general | |

| | |
|---|---|
| ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs | |

| | |
|---|---|
| followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| |
|---|
| Local exhaust ventilation and / or general ventilation are / is advisable. |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. | |

| | |
|---|---|
| Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC19: Manual activities involving hand contact<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |

| | |
|---|---|
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual | |

| | |
|---|---|
| phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |

| ***Exposure estimate and reference to its source*** | |
|---|---|
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 11. Short title of exposure scenario

Use in food products, (use in professional settings)
PW; SU4; ERC8b; PROC3, PROC4, PROC5, PROC8a, PROC8b, PROC9, PROC10, PROC11, PROC13, PROC19, PROC26

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8b: Widespread use of reactive processing aid (no inclusion into or onto article, indoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are | |

| | |
|---|---|
| implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: professional |

| | |
|---|---|
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable | |

| | |
|---|---|
| working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure | |

| | |
|---|---|
| containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8a: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities<br>Use domain: professional |
| | Additional PROC(s) covered: Transfer of substance or mixture (charging and discharging) at dedicated facilities Transfer of substance or preparation into small containers (dedicated filling line, including weighing). |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with | |

| | |
|---|---|
| contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |

| | |
|---|---|
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |

| | |
|---|---|
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC11: Non industrial spraying<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide | |

| | |
|---|---|
| specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring.<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. | |

| | |
|---|---|
| Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |

| | |
|---|---|
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC19: Manual activities involving hand contact<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through | |

| | |
|---|---|
| time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss | |

| | |
|---|---|
| events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 12. Short title of exposure scenario

Use in/as Photochemicals, (use in professional settings)

PW; SU6b, SU7; ERC8e; PROC3, PROC4, PROC5, PROC8b, PROC8a, PROC9, PROC10, PROC11, PROC13, PROC15, PROC19, PROC26

## Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8e: Widespread use of reactive processing aid (no inclusion into or onto article, outdoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air | |

| | |
|---|---|
| changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs | |

| | |
|---|---|
| followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| |
|---|
| Local exhaust ventilation and / or general ventilation are / is advisable. |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. | |

| | |
|---|---|
| Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC8b: Transfer of substance or mixture (charging and discharging) at dedicated facilities<br>Use domain: professional |
| | Additional PROC(s) covered: Transfer of substance or mixture (charging and discharging) at non-dedicated facilities Transfer of substance or preparation into small containers (dedicated filling line, including weighing). |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular | |

| | |
|---|---|
| inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC10: Roller application or brushing<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance | 0.0001 Pa |

| | |
|---|---|
| during use | |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |

| | |
|---|---|
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC11: Non industrial spraying<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |

| | |
|---|---|
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC13: Treatment of articles by dipping and pouring.<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. | |

| | |
|---|---|
| Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: professional |

| | |
|---|---|
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable | |

| | |
|---|---|
| working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC19: Manual activities involving hand contact<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure | |

| | |
|---|---|
| containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: professional |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize | |

| | |
|---|---|
| exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********

## 13. Short title of exposure scenario

Use in/as Fertilizers, (liquid products), (consumer use)
C; ERC8e; PC12

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8e: Widespread use of reactive processing aid (no inclusion into or onto article, outdoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

**********

## 14. Short title of exposure scenario

Use in/as Photochemicals, (consumer use)
C; ERC8e; PC30

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | ERC8e: Widespread use of reactive processing aid (no inclusion into or onto article, outdoor)<br>As no environmental hazard was identified no environmental-related exposure assessment and risk characterization was performed. |
| ***Operational conditions*** | |

**********

## 15. Short title of exposure scenario

Formulation & (re)packing of substances and mixtures
IS; PROC3, PROC4, PROC5, PROC15, PROC26, PROC28

### Control of exposure and risk management measures

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC3: Manufacture or formulation in the chemical industry in closed batch processes with occasional controlled exposure or processes with equivalent containment condition<br>Use domain: industrial |
| ***Operational conditions*** | |

| | |
|---|---|
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable | |

| | |
|---|---|
| chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC4: Chemical production where opportunity for exposure arises<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure | |

| | |
|---|---|
| containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC5: Mixing or blending in batch processes<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid | |

| | |
|---|---|
| splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| |
|---|
| **Contributing exposure scenario** |

| | |
|---|---|
| **Use descriptors covered** | PROC15: Use a laboratory reagent.<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear | |

| | |
|---|---|
| suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source | |

| | |
|---|---|
| Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

| | |
|---|---|
| **Contributing exposure scenario** | |
| **Use descriptors covered** | PROC26: Handling of solid inorganic substances at ambient temperature<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in | |

| | |
|---|---|
| place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |

| ***Additional good practice advice*** |
|---|
| Local exhaust ventilation and / or general ventilation are / is advisable. |
| Local exhaust ventilation and / or general ventilation are / is advisable. |

| **Contributing exposure scenario** | |
|---|---|
| **Use descriptors covered** | PROC28: Manual maintenance (cleaning and repair) of machinery<br>Use domain: industrial |
| ***Operational conditions*** | |
| Physical state | liquid |
| Vapour pressure of the substance during use | 0.0001 Pa |
| ***Risk Management Measures*** | |
| Minimise number of staff exposed. Ensure segregation of worker from the source Ensure minimization of manual phases Avoid contact with contaminated tools. Supervision in place to check that the RMMs in place are being used correctly and OCs followed. Provide basic employee training to prevent/minimize exposures. Avoid skin contact. Avoid splashing. Clean equipment and the work area every day. Clean up contamination as soon as they occur. Ensure good work practices are implemented. Ensure that near miss events are documented. Regular inspection and maintenance of equipment and machines. Restrict access to authorised persons. Provide specific employee training to prevent/minimize exposures. Wash off any skin contamination immediately. | |
| Containment as appropriate Ensure containment of the emission source Provide a good standard of general ventilation (not less than 3 - 5 air changes per hour) Handle substance within closed system. Ensure containment of the emmision source and provide extract ventilation to points where emission occur | |
| Regular cleaning of equipment and work area., Change gloves, if duration of activity exceeds break through time, Wear suitable face shield, Avoid inhalation of the product. | |
| Avoid contact with eyes. | |

| | |
|---|---|
| Use suitable eye protection. | |
| Avoid contact with contaminated tools. Avoid skin contact. Avoid splashing. Wash off any skin contamination immediately. | |
| Wear suitable face shield, Wear suitable coveralls to prevent exposure to the skin., Use suitable chemically resistant gloves., Wear suitable working clothes., Wear suitable chemically resistant gloves with sleeves. | |
| Ensure minimization of manual phases Ensure that no inhalable dusts are generated. | |
| Avoid inhalation of the product., Wear suitable respiratory protection. | |
| ***Exposure estimate and reference to its source*** | |
| Assessment method | Qualitative assessment |
| ***Additional good practice advice*** | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |
| Local exhaust ventilation and / or general ventilation are / is advisable. | |

**********